

Physics 201

Fall 2026

Office: Luter 304, or online

Office Hours: MWF 1:00-2:00pm, or by appointment - in person or on Google Meet

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594-7451

GENERAL PHYSICS

Text: OpenStax University Physics – Volume I – <http://www.openstax.org>

This textbook is freely available online, and there is a print version available on Amazon that is quite reasonably priced. There is also an OpenStax app for your phone or tablet that I highly recommend. The homework problems will be from this book as well.

Course Learning Objectives:

Upon completion of PHYS201, a student should be able to:

- 1) identify and apply appropriate models, laws and equations to physical problems involving forces and motion.
- 2) differentiate between scalar and vector quantities.
- 3) apply free body diagrams and unit vector notation to solve problems involving vectors in translational and rotational kinematics and dynamics.
- 4) solve problems in mechanics using conserved quantities.

Physics Topics: Units, conversion, and problem solving. Kinematics (position, velocity, acceleration). Circular motion. Forces and Newton's laws. Work, kinetic energy, and potential energy. Linear momentum and collisions. Rotational kinematics and dynamics.

Homework: Assignments will be given each week and will be due the following week. Homework will not be turned in but will instead be submitted via the WebAssign web portal.

TO RECEIVE HOMEWORK CREDIT, YOU MUST SUBMIT YOUR ANSWERS VIA THE WEBASSIGN PORTAL.

You can find the homework assignments, as well as other important resources, at:

<http://www.webassign.net/>

You will enroll with a class key: cnu 3861 2811

You can also find lots of interesting things on my GitHub repository for this course, at:

<https://github.com/brash99/phys201>

Exams: Two hourly exams and a comprehensive final.

Grading:	Homework	25%
	Each hourly exam	20% x 2 = 40%
	Final exam	35%

Homework is due by MIDNIGHT on the due date. If you do not complete the web-based assignment by this time, the system will NOT allow you to submit the assignment. Normally, no extensions will be given, except with a valid documented excuse (typically medical reasons or other university sanctioned activities).

If you miss one of the hourly exams because of a valid, documented reason, the grade portion for that exam will be added to your final exam weighting. No make-up exams will be given under any circumstances.

The evaluation of your performance in this course will be based entirely on the regular homework assignments and scheduled exams. There is no possibility to do extra work for extra credit.

Final grades will be assigned as follows:

A = 87-100%; A- = 80-86%
B+ = 77-79%; B = 73-76%; B- = 70-72%
C+ = 67-69%; C = 63-66%; C- = 60-62%
D+ = 57-59%; D = 53-56%; D- = 50-52%; F < 50%

Expectations:

Physics is a mathematically intensive subject that presents everyone with occasional (and sometimes frequent) difficulties. You are expected to encounter problems that will challenge your limits of understanding. As a general rule, a typical student should expect to put in 5-6 hours per week outside of class to reach his/her potential in this course (though this will vary for each person, depending on abilities and background). Always contact your instructor if you need help and/or feel overwhelmed - I wish to see each and every student succeed in this class and at CNU in general!

INW Course Objectives:

This class is a lecture course that is certified to fulfill part of the Investigating the Natural World (INW) requirement in the Liberal Learning Curriculum. The learning objectives for INW lecture courses are:

- Identify the methods of inquiry that lead to scientific knowledge.

We will be discussing the historical development of many of the concepts that will be presented allowing you to observe how these principles were elucidated.

- Distinguish science from pseudo-science.

All scientific theories must make predictions and those predictions must be verified by experiment. We will touch upon some non-theories which were presented as explanations for observed effects.

- Make predictions about natural phenomena using theories and models as unifying principles.

As we develop the theories to be presented, we'll investigate the consequences of those theories and verify that the implied predictions are witnessed in nature.

- Discriminate between association and causation, and identify the types of evidence used to establish causation

A mathematical formula only associates two phenomena, we will discuss the experiments that helped determine that one actually caused the other.

Honor Code and Cheating

As all of you know, CNU has an Honor Code, which we all agreed to by both signing and giving a pledge. The spirit of that Honor Code will be strictly observed in this course.

With that said, we need to have a longer discussion in about what "violating the honor code" actually looks like in this type of course.

Collaboration and the thoughtful use of outside resources are encouraged in this course. You may discuss concepts and problem-solving strategies with classmates and may use textbooks, websites, or AI tools such as ChatGPT to help explain ideas or evaluate your reasoning. When such assistance contributes materially to your work, you should acknowledge it. However, the work you submit must reflect your own understanding: you must personally carry out, understand, and be able to explain each solution. Copying or submitting a solution produced by another person or tool does not become acceptable merely because the source is acknowledged. Exams must be completed without unauthorized assistance.

Additional Information, provided by the CNU Administration, which you may find useful (and with which I happen to agree, in general):

**Christopher Newport University
Official Syllabi Statements
AY 2026-2027**

Academic Honesty:

Academic honesty is fundamental to the mission of a university. Cheating, plagiarism, and any other violations of the Honor Code may result in academic remedies such as a failing grade on the assignment or the course (depending on the severity of the offense). Violations will also be reported to the Center for Honor Enrichment and Community Standards (CHECS).

The Christopher Newport University Honor Code:

“On my honor, I will maintain the highest standards of honesty, integrity and personal responsibility. This means I will not lie, cheat or steal, and as a member of this academic community, I am committed to creating an environment of respect and mutual trust.”

University Statement on Inclusion and Belonging:

The Christopher Newport University community engages and respects different viewpoints, understands the cultural and structural context in which those viewpoints emerge, and questions the development of our own perspectives and values, as these are among the fundamental tenets of a liberal arts education.

Accordingly, we affirm our commitment to a campus culture that embraces the full spectrum of human attributes, perspectives, and disciplines and offers every member of the University the opportunity to become their best self.

Understanding and respecting differences can best develop in a community where members learn, live, work, and serve among individuals with diverse worldviews, identities, and values. We are dedicated to upholding the dignity and worth of all members of this academic community, such that all may engage effectively and compassionately in a pluralistic society.

If you have specific questions, suggestions, or concerns regarding inclusion on campus please contact belonging@cnu.edu.

Disabilities/Accessibility:

In order for a student to receive an accommodation due to a disability, that disability must be on record in the Office of Student Affairs, 3rd Floor, David Student Union (DSU). If you have a diagnosed disability, please contact Accessibility & Care Team Support in Student Affairs (594-7160) or email careteam@cnu.edu to discuss your needs. You may visit <https://cnu.edu/life/acts/accessibility/> to request accommodations or view additional information.

In order to provide access to educational programs in the best manner possible, students with documented disabilities should notify the instructor at least seven days prior to the

point at which they require an accommodation (the first day of class is recommended), in private, if an accommodation is needed. The instructor will provide students with disabilities with the reasonable accommodations approved and directed by the Office of Accessibility and Care Team Support. Work completed before the student notifies the instructor of his/her disability may be counted toward the final grade at the sole discretion of the instructor.

Success:

I want you to succeed in this course and at Christopher Newport. I encourage you to contact me during office hours or schedule an appointment to discuss course content or ask questions. If I become concerned about your course performance, attendance, engagement, or well-being, I will contact you first. I also may submit a referral through our Captains Care Program. The referral will be received by Student Affairs, the Center for Academic Success, and other departments, when appropriate. If you are an athlete, the Director of Student-Athlete Transition and Support will be notified. Someone from Student Affairs or the Center for Academic Success will contact you to set up a plan for success and connect you with the appropriate resources. Please remember that this is a means for me to support you and help foster your success at Christopher Newport.

Academic Support:

The Center for Academic Success (CAS) provides numerous free resources to help students succeed in their courses. These resources include individual peer mentoring, peer tutoring, and group/test review sessions. Academic Success Coordinators can meet with you regularly to discuss your plan to complete your coursework while balancing your extracurricular activities. They can help you develop individualized study skills, create time-management plans, and techniques to prepare for exams and other assignments. The Center also houses the Alice F. Randall Writing Center. Writing consultants can help you at any stage of the writing process, from invention to development of ideas to polishing a final draft.

You may contact the Center for Academic Success to request a tutor, meet with a peer mentor or writing consultant, or make an appointment to talk with a staff member about study skills and strategies. The Center is located in Christopher Newport Hall, first floor, room 124. You may email academicsuccess@cnu.edu, call (757) 594-7684, or visit <https://cnu.edu/center-for-academic-success>

Public Health

The university will provide guidance on public health issues and students will be expected to comply with university protocols.

Course Materials:

All content created and assembled by the faculty member and used in this course is to be considered intellectual property owned by the faculty member and Christopher Newport University. It is provided solely for the private use of the students currently enrolled in this course. To ensure the free and open discussion of ideas, students may not make any original course content, including but not limited to lectures, discussions, videos, handouts, and/or activities, available to anyone not currently enrolled in the course

without the instructor's advance written permission. This means that students may not record, download, screenshot, or in any way copy original course material for the purpose of distribution beyond this course. A violation may be considered theft. It is the student's responsibility to protect course materials when accessing them outside the physical classroom.